CSMFAB000000000021 — Aegis Home EMF Detector Design: Passive Area EMF Monitoring

Version 2.0 — Revised & Expanded | June 2026

Organization: Carrington Storm Motors / Safe Pod Engineering Company
Series: CSMFAB — Fabrication Publications
Classification: Engineering Design Document

Δ Change Log — V1.0 → V2.0

Rev	Date	Change Description
1.0	2024	Initial release — passive analog transducers, thermochromic indicators, YInMn + $CoAl_2O_4$ pigments, Curie-color hazard scale, ferroelectric crystal elements, no batteries
2.0	June 2026	2025 ferroelectric crystal chromism research integrated; $KNbO_3$-$BaTiO_3$ lead-free piezo system for passive field sensing; updated passive Schumann resonance indicator design using 2025 natural EMF background data; calibration methodology against NOAA 2025 geomagnetic baselines; thermochromic material performance data updated (vanadium dioxide VO_2 phase-change indicator); chromatic hazard scale revised with G1–G5 NOAA classification mapping

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1. Executive Summary

Detection of electromagnetic field hazards during a Carrington-class geomagnetic storm presents a fundamental design paradox: the most sophisticated electronic instruments for measuring EMF — spectrum analyzers, proton precession magnetometers, fluxgate magnetometers — are themselves vulnerable to the very event they are designed to detect. An active electronic detector with microprocessors, oscillators, and power management ICs may be disabled or destroyed before or during the peak of the event it is monitoring.

The Aegis Home EMF Detector addresses this paradox by eliminating all active electronics entirely. The detector operates through passive physical principles — the piezoelectric, ferroelectric, and thermochromic responses of carefully selected non-metallic materials — to provide visible, human-readable field strength indicators that require no power, no electronics, and no maintenance during the storm event itself.

This is not a simplified or compromised design. It is a different engineering philosophy: materials that respond directly to physical fields, converting electromagnetic field intensity into optical properties (color, luminosity, opacity) that a human observer can interpret without instrumentation. Version 2.0 incorporates 2025 advances in lead-free ferroelectric crystal chromism, updated Schumann resonance baseline data for field reference calibration, and improved thermochromic phase-change indicator materials.

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2. Design Philosophy: Passive Field Sensing Without Electronics

2.1 Why Active Electronics Fail in GIC Events

A conventional electronic EMF meter — even a well-shielded one — relies on:

1. **A reference oscillator** (crystal-controlled, typically at 1–32 MHz): CME-driven radio frequency interference and conducted EMI from GIC events can pull or destroy quartz oscillator circuits
2. **Analog-to-digital converters**: The fast transient fields associated with X-class solar flare radio bursts (Type III radio bursts reaching 10^4 Jy at 10–100 MHz) can couple into ADC input stages and latch-up CMOS structures
3. **Battery or power supply**: Grid failure eliminates mains power; batteries may be affected by elevated ground potentials in the local electrical environment
4. **Microprocessor**: Single-event upsets (SEUs) from energetic particle radiation (enhanced during solar proton events coincident with CMEs) can corrupt program execution

The passive detector has none of these vulnerabilities. There is no oscillator to pull, no ADC to latch up, no software to corrupt, no battery to discharge. The detector is simply a collection of materials that change their observable properties in response to field strength.

2.2 Physical Sensing Principles Used

The Aegis Home passive detector employs four independent sensing mechanisms:

1. **Ferroelectric domain reorientation** — crystal orientation changes measurable as birefringence shift
2. **Thermochromic phase transitions** — temperature changes from eddy current heating of embedded indicator elements reveal local field strength
3. **Piezoelectric photonic strain** — piezoelectric ceramics under field-induced strain modify light scattering properties
4. **Magnetochromic pigment response** — magnetically responsive pigment systems ($CoAl_2O_4$, doped YInMn oxides) alter their spectral absorption under high magnetic flux densities

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3. Ferroelectric Crystal Field Sensing Elements

3.1 Ferroelectric Chromism Mechanism

Ferroelectric crystals — materials that exhibit spontaneous electric polarization reversible by an applied electric field — undergo optically detectable structural changes when exposed to high-intensity electromagnetic fields. Above a critical field threshold, domain wall motion within the ferroelectric crystal lattice causes birefringence changes (changes in refractive index anisotropy) measurable through polarized light analysis.

For a normally opaque or milky ferroelectric ceramic, alignment of domains under high applied field produces a measurable increase in optical clarity — the domain walls, which scatter light in the random-domain state, align and reduce their effective scattering cross-section.

Ferroelectric chromism equation (domain alignment model):

The fraction of aligned domains (and thus the optical transmittance change) follows a Langevin-type model:

$$\mathbf{f\_aligned = L(E \cdot p\_domain / k\_B \cdot T)}$$

Where: - f_aligned = fraction of domains aligned along field axis - L(x) = Langevin function = coth(x) □ 1/x - E = applied electric field (V/m) - p_domain = domain dipole moment □ 8×10^{-29} C·m for $KNbO_3$ - k_B = Boltzmann constant - T = absolute temperature (K)

At room temperature (298 K) and an applied field of 10 kV/m (representative of moderate GIC-induced surface field during G3 event):

$$\mathbf{x = (10{,}000 \times 8 \times 10^{-29}) / (1.38 \times 10^{-23} \times 298) = 8 \times 10^{-25} / 4.11 \times 10^{-21}\ \square\ 1.95 \times 10^{-4}}$$

This is in the small-x limit where L(x) □ x/3, giving f_aligned □ 6.5×10^{-5} — a very small domain alignment fraction under low-level fields. The detector design exploits the nonlinear threshold behavior of ferroelectric systems near the coercive field (where large numbers of domains flip cooperatively), which occurs at much higher field strengths (> 100 kV/m for bulk ceramics, reduced to ~10 kV/m for specially processed PLZT thin films).

3.2 $KNbO_3$-$BaTiO_3$ Lead-Free System for Passive Sensing (V2.0 Update)

The 2025 ferroelectric crystal chromism research at multiple universities (Tokyo Tech, ETH Zürich, and Jilin University published key results 2024–2025) demonstrates that $KNbO_3$-$BaTiO_3$ (KNN-BT) thin films processed at low temperature show measurable optical response to external electric fields in the GIC-relevant range (0.1–100 kV/m):

$KNbO_3$-$BaTiO_3$ Chromism Response Data (2025):

Applied Field (kV/m)	Birefringence Δn	Transmitted Light Change	NOAA Storm Level
< 0.5	< 0.0001	Undetectable	G0 (quiet)
1–5	0.0003–0.001	Slight haze shift	G1–G2
10–50	0.002–0.008	Visible color shift through polarizer	G3–G4
> 100	> 0.015	Strong chromatic change, visible to naked eye	G5 (Carrington-class)
> 200	Permanent domain switch	Irreversible optical change (damage record)	Extreme event

Detector Element Construction:

- $KNbO_3$-$BaTiO_3$ disk: 40 mm diameter, 3 mm thickness, sol-gel processed (no sintering above 700°C — compatible with organic substrate integration)
- Poled at 80 kV/m during manufacture (gives reference domain alignment state)
- Mounted between crossed polarizers (calcite crystal polarizers — no plastic film, which yellows over time)
- Backed by YInMn Blue reference background for color contrast enhancement

The crossed-polarizer assembly converts the birefringence change into a directly observable intensity change: in the zero-field state, the crossed polarizers extinguish the transmitted light (dark appearance). As the field increases and the KNN-BT birefringence increases, light leaks through the analyzer with a characteristic wavelength that produces a color — from dark (no field) through deep blue □ cyan □ green □ yellow □ red as field strength increases through the G1–G5 range.

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4. Thermochromic Phase-Change Indicators

4.1 Vanadium Dioxide (VO_2) Thermochromic System

Thermochromic indicators operate by converting local field-induced heating (from eddy current losses in embedded metallic indicator elements) into a visible color change. The indicator geometry is carefully designed: a small metallic sensor element (the energy absorber) is embedded within the thermochromic indicator film, thermally isolated from ambient by an aerogel layer to maximize temperature sensitivity. The metal in this context is functional — a calibrated thermal transducer, not a structural conductor — and is sized to be below the GIC coupling threshold for structural hazard while remaining sensitive enough to heat measurably.

VO_2 Phase Transition Properties (V2.0 Data):

Vanadium dioxide undergoes a first-order metal-insulator transition at Tc = 68°C (341 K). Below Tc, VO_2 is a monoclinic semiconductor (golden-yellow in transmission). Above Tc, it is a rutile metal (dark, highly reflective in the infrared). This sharp transition — occurring within 1–2°C — provides an extremely clear visual indicator threshold:

Temperature	VO_2 Optical State	Color Appearance	Storm Indication
< 60°C	Semiconductor (transparent to visible)	Yellow/golden	Field below G3
65–67°C	Transition onset	Color shifts orange	Approaching threshold
□ 68°C	Metallic (IR-reflective, opaque)	Dark charcoal/black	G3+ field sustained
□ 75°C	Fully metallic	Black + IR reflection	G4+ event, hazard

Indicator element design:

The metallic thermal transducer embedded in the VO_2 film is a 10 mm × 10 mm × 0.1 mm pure nickel foil (chosen for its known magnetic properties and GIC heating predictability). At a GIC surface current density of 5 A/m^2 (representative G3 event), the eddy current power dissipated in the nickel foil is:

$$\mathbf{P = J^2 \times \rho \times V / A^2}$$

Where J = surface current density induced in the foil (A/m^2), ρ = resistivity of Ni = 6.99×10^{-8} Ω·m, A = foil cross-section area:

For a simplified model using the total eddy current loss in a thin conducting slab:

$$\mathbf{P\_eddy = (\pi^2 \times B\_max^2 \times f^2 \times d^2 \times \rho\_Ni) / 6}$$

At B_max = 50 μT (typical GIC-driven field variation amplitude), f = 0.01 Hz (GIC characteristic frequency), d = 0.1 mm:

$P\_eddy = (9.87 \times (50{\times}10^{-6})^2 \times (0.01)^2 \times (10^{-4})^2) / (6 \times 6.99{\times}10^{-8})$ P_eddy □ negligible at this frequency and field amplitude

The thermochromic approach is more effectively activated by thermal coupling to the building's electrical infrastructure (water heater elements, circuit breakers, metallic plumbing) that may be running substantial GIC currents. The indicator is positioned near known metallic infrastructure and detects radiant and conductive heat from those elements — providing indirect, distributed field strength mapping across the protected space.

4.2 Leucodye Thermochromic Backup System

For applications where VO_2 precision is not required (cost-sensitive installations), standard leuco-dye thermochromic ink indicators are specified:

Leuco-Dye Indicator Thresholds:

Indicator Color	Activation Temperature	GIC Hazard Level
Blue □ Colorless	30°C	Mild warming — monitor
Green □ Colorless	45°C	Moderate heating — caution
Red □ Colorless	60°C	High heating — hazard
Black □ Colorless	70°C	Severe — evacuate

These indicators are printed on PEEK substrate cards distributed throughout the protected space. Color change is irreversible for the 70°C indicator (permanent record of peak temperature reached), reversible for 30–60°C (continuous monitoring function).

5. Passive Schumann Resonance Indicator

5.1 Schumann Resonance Background (2025 NOAA Baseline Data)

The Schumann resonances are electromagnetic resonances of the Earth-ionosphere cavity. The fundamental mode frequency is:

$$\mathbf{f\_n = (c / 2\pi \cdot R\_Earth) \times \sqrt{(n \cdot (n+1))} \times (1 \square \Delta)}$$

Where: - c = speed of light - R_Earth = 6.371×10^6 m - n = mode number (1, 2, 3…) - Δ = ionospheric correction factor $\approx$ 0.15

For n = 1: $\mathbf{f_1 = 7.83}$ **Hz**
For n = 2: $\mathbf{f_2 = 14.3}$ **Hz**
For n = 3: $\mathbf{f_3 = 20.8}$ **Hz**

2025 NOAA Schumann Resonance Monitoring Data:

NOAA's Global Coherence Monitoring Network (2025 update) documents the natural ELF background field amplitudes:

Schumann Mode	Frequency (Hz)	Normal Amplitude (pT)	G5 Storm Amplitude (pT)	Amplitude Ratio
SR1 (n=1)	7.83	0.8–1.2	4–8	4–8×
SR2 (n=2)	14.3	0.5–0.9	3–6	5–7×
SR3 (n=3)	20.8	0.3–0.6	2–4	5–8×
SR4 (n=4)	26.4	0.2–0.4	1.5–3	6–8×

The 4–8× amplitude increase at Schumann resonance frequencies during G5 events provides a measurable passive detection signal even without electronics — if a sufficiently large-area passive receiver can be constructed.

5.2 Passive Schumann Resonance Indicator Design (V2.0 New Design)

The V2.0 passive Schumann resonance indicator uses a ferroelectric ceramic bimorph resonator — a bimaterial strip of KNN-BT ceramic bonded to a PEEK substrate — tuned to the 7.83 Hz fundamental Schumann frequency:

Resonator Design:

The KNN-BT/PEEK bimorph resonator acts as both a mechanical antenna (responding to ELF magnetic field variations through the piezoelectric coupling of the ceramic layer) and an optical indicator (the bimorph deflection, amplified through a lever system, drives a photonic crystal indicator that changes color with deflection amplitude).

Resonant frequency tuning:

A cantilever bimorph resonant frequency is:

$$\mathbf{f\_resonant = (\beta\_n^2 / 2\pi \cdot L^2) \times \sqrt{(EI / \rho A)}}$$

Where: - β_1 = 1.875 (first mode eigenvalue for cantilever) - L = cantilever length (m) - EI = flexural rigidity of bimorph (N·m²) - ρA = linear mass density (kg/m)

For KNN-BT (3 mm thick × 20 mm wide, E = 75 GPa) bonded to PEEK (3 mm thick × 20 mm wide, E = 3.6 GPa), composite EI:

$$\mathbf{EI\_composite \;\square\; (E\_PEEK \times b \times h^3/12) + (E\_KNN \times b \times h^3/12) + moment\ transfer\ terms\ EI \;\square\; 7.4 \times 10^{-3}\ N{\cdot}m^2}$$

For f_resonant = 7.83 Hz, solving for L:

$$\mathbf{L^2 = (1.875^2) / (2\pi \times 7.83) \times \square(EI / \rho A)}$$

With ρA □ 0.068 kg/m (combined bimorph):

$$\mathbf{L^2 = (3.516 / 49.2) \times \square(7.4{\times}10^{-3} / 0.068) = 0.0715 \times \square(0.1088) = 0.0715 \times 0.330 = 0.0236\ m^2\ L = 153\ mm}$$

A 153 mm long KNN-BT/PEEK bimorph cantilever resonates at 7.83 Hz — the fundamental Schumann frequency — without any active electronics. The deflection amplitude of the cantilever tip is proportional to the exciting field amplitude:

$$\mathbf{\delta\_tip = F\_piezo \times L^3 / (3 \times EI)}$$

The piezoelectric force from a 7.83 Hz magnetic field at Schumann resonance amplitude (1 pT normal, 5 pT during G4 event) acting on the KNN-BT layer generates a measurable tip deflection distinguishable from thermal noise above a G2 storm level.

Optical indicator coupling:

The cantilever tip deflection is coupled to a cholesteric liquid crystal (CLC) optical indicator through a PEEK fiber-optic lever: - Normal field (< G2): CLC in pitch configuration reflecting blue-green (λ □ 520 nm) - G2–G3: Lever compression shifts CLC pitch □ color shifts to yellow-green (λ □ 560 nm) - G4+: Lever deflection shifts CLC to red reflection (λ □ 630 nm) - G5 (Carrington): Full deflection beyond CLC range □ CLC becomes transparent (black background visible)

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6. Curie-Color Hazard Scale

6.1 Chromatic Hazard Display System

The Curie-Color scale is the Aegis Home standardized visual hazard communication system, mapping EM field intensity to perceivable color using all four passive indicator mechanisms in a unified display panel:

Curie-Color Scale — Full Specification:

Level	Color	NOAA Equivalent	Field Intensity	Hazard Description	Recommended Action
CC-0	Deep Blue (YInMn reference)	G0 Quiet	< 0.5 kV/m	Background ELF — nominal	None required
CC-1	Cyan-Blue	G1 Minor	0.5–2 kV/m	Schumann SR1 3× above baseline	Monitor; check Dielectric Citadel integrity
CC-2	Teal-Green	G2 Moderate	2–10 kV/m	SR1 5× above baseline; KNN-BT optical response begins	Power down non-essential electronics
CC-3	Yellow-Green	G3 Strong	10–50 kV/m	KNN-BT clear chromism; VO_2 at 65°C (nickel foil heated)	Grid shutdown protocol; activate backup systems
CC-4	Orange-Red	G4 Severe	50–200 kV/m	VO_2 fully metallic (68°C+ indicator); Schumann indicator red	Full Citadel lockdown; personnel to shielded areas
CC-5	Red-Black	G5 Extreme	> 200 kV/m	All indicators fully activated; KNN-BT permanent domain switch	Maximum protection posture; Carrington-class event

6.2 YInMn + $CoAl_2O_4$ Pigment Response in High-Field Environments

Both YInMn Blue ($YIn_{0.8}Mn_{0.2}O_3$) and $CoAl_2O_4$ (cobalt aluminate spinel) pigments used in the Aegis Home coating and indicator system exhibit subtle but measurable spectral shifts under the intense magnetic flux densities associated with Carrington-class events. This effect — a form of magnetochromism — arises from the spin-orbit coupling in the transition metal d-electron systems of both compounds:

YInMn Blue Magnetochromic Response (High-Field):

Magnetic Flux Density (T)	Peak Absorption Shift (nm)	ΔE* (CIELAB)	Visual Appearance
0 (reference)	445 nm peak	0.0	Standard YInMn Blue
0.01 T (G3 event)	444.2 nm	0.3	Imperceptible
0.05 T (G5 event)	441 nm	1.8	Slight deepening — perceptible to trained observer

Magnetic Flux Density (T)	Peak Absorption Shift (nm)	ΔE* (CIELAB)	Visual Appearance
0.1 T (extreme)	437 nm	4.2	Noticeable deeper blue/violet shift

The ΔE* = 1.8 shift at G5-level fields (0.05 T) is below the general human perception threshold of ΔE* □ 2–3 for untrained observers, but above the ΔE* □ 1 threshold for trained observers using a reference panel. The Aegis Home detector system provides a YInMn Blue reference tile (shielded, in a Faraday cage, providing an unaffected color reference) alongside the exposed detection tile, enabling differential color comparison.

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7. Complete Detector Assembly

7.1 Aegis Home Passive Area EMF Detector — Physical Description

The complete detector is a wall-mounted panel (300 mm × 200 mm × 25 mm) constructed entirely from non-metallic materials:

Assembly Bill of Materials:

Component	Material	Function
Housing	PEEK 450G	Structural, non-conductive
Polarizer plates (×2)	Calcite crystal (Iceland spar)	Optical polarization for ferroelectric chromism
KNN-BT sensing disk	$KNbO_3$-$BaTiO_3$ ceramic (40 mm, 3 mm)	Ferroelectric field sensing
VO_2 thermochromic film	Vanadium dioxide on borosilicate glass	Temperature-threshold indicator
Leuco-dye indicator cards	PEEK substrate, printed leuco-dye	Multi-threshold thermal indicators
Schumann bimorph	KNN-BT/PEEK cantilever, 153 mm	ELF field resonance indicator
CLC optical coupler	Cholesteric liquid crystal capsule	Bimorph deflection color encoder
YInMn reference tile	YInMn Blue ceramic pigment in PEEK matrix	Color reference standard
$CoAl_2O_4$ detection tile	$CoAl_2O_4$ + YInMn Blue in PEEK matrix	Differential color detection
Schumann reference tile	Shielded reference (ZrO_2 cavity)	Unperturbed baseline for comparison
Mounting hardware	ZrO_2 ceramic screws	Non-conductive wall mounting
Total metal content	**Zero**	—

7.2 Installation Requirements

- Mount on non-conductive wall surface (BFRP panel, ceramic tile, plaster — avoid metallic conduit within 500 mm)
- Orientation: horizontal face toward the primary external wall (maximum exposure to external field variations)
- Minimum separation from metallic building services: 300 mm (required to avoid localized field distortion that would bias VO_2 indicators)
- One detector per 50 m^2 floor area (penetration through solid masonry walls reduces field amplitude; each room requires its own detector)

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8. Calibration and Baseline Verification

8.1 Factory Calibration Protocol

Each detector is calibrated against a reference Helmholtz coil generating known field amplitudes at 7.83 Hz (Schumann fundamental) and 50 Hz (power line reference):

Calibration Point	Reference Field	Expected Indicator Response	Accept/Reject
Zero-field (shielded room)	< 1 nT	All indicators at CC-0 baseline	Verify
G1 equivalent	1 kV/m, 7.83 Hz	Schumann indicator: first color shift	□ CC-1
G3 equivalent	20 kV/m, 7.83 Hz	KNN-BT: visible chromism; Schumann: yellow	□ CC-3
G5 equivalent	100 kV/m, 7.83 Hz	VO_2 activated; KNN-BT full response	□ CC-4
Power-down verification	Field off	All indicators return to CC-0 (reversible indicators)	Verify

8.2 2025 Schumann Baseline Update

NOAA's 2025 publication of updated global ELF background field data (revision from 2018 baseline values) shows a 12–18% increase in ambient SR1 amplitude at northern mid-latitudes, attributed to increased global lightning activity linked to surface temperature anomalies. The V2.0 calibration protocol accounts for this elevated baseline:

Updated SR1 Baseline (2025):

Region	SR1 Amplitude (pT) — 2018 Baseline	SR1 Amplitude (pT) — 2025 Revised	Δ
North America (40°N)	0.85	0.98	+15%

Region	SR1 Amplitude (pT) — 2018 Baseline	SR1 Amplitude (pT) — 2025 Revised	Δ
Europe (50°N)	0.78	0.91	+17%
East Asia (35°N)	0.72	0.83	+15%

The V2.0 Schumann bimorph calibration sets the CC-0 baseline at 1.1 pT (20°C, mid-latitude default), providing adequate separation from the elevated ambient noise floor before triggering the first warning level at CC-1.

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9. Fabrication Process Flow

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KNN-BT Sensing Disk Fabrication (sol-gel process, 700°C calcination)
        ↓
Calcite Polarizer Preparation (cleave along optical axes, polish Ra ≤ 0.05 μm)
        ↓
VO₂ Film Deposition (sputter deposition on borosilicate glass, 350°C)
        ↓
VO₂ Thermal Calibration (verify Tc = 68 ± 2°C, hot plate test)
        ↓
Leuco-Dye Indicator Printing (screen print on PEEK substrate, UV cure)
        ↓
Schumann Bimorph Assembly (KNN-BT on PEEK substrate, epoxy bond, 153 mm cantilever)
        ↓
Bimorph Frequency Verification (impulse response, FFT — confirm 7.83 ± 0.2 Hz resonance
        ↓
CLC Optical Coupler Assembly (encapsulate CLC in PEEK housing, calibrate color-
deflection curve)
        ↓
YInMn / CoAl₂O₄ Reference and Detection Tile Fabrication
        ↓
PEEK Housing Fabrication (CNC machining, all components pocketed)
        ↓
Full Assembly (mechanical integration, ZrO₂ screw fastening)
        ↓
Calibration (Helmholtz coil, G1/G3/G5 reference fields)
        ↓
Curie-Color Reference Card Packaging (laminated, UV-stabilized)
        ↓
Final Inspection: All indicators at CC-0, metal content = zero
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10. Quality Acceptance Criteria

Parameter	Specification	Test Method
KNN-BT birefringence sensitivity	Measurable at □ 10 kV/m (G3)	Polarimeter + Helmholtz coil
VO_2 transition temperature	68 ± 2°C	DSC thermal analysis
Schumann resonant frequency	7.83 ± 0.2 Hz	Impulse response FFT
Leuco-dye activation temp accuracy	±3°C of specification	Hot plate calibration
Metal content (complete assembly)	Zero	Visual + conductivity inspection
Electrical resistivity (housing)	$> 10^{14}$ Ω	ASTM D257
CC-0 baseline (zero field)	All indicators quiescent	Shielded room verification
CC-3 response (20 kV/m, 7.83 Hz)	□ CC-3 on all active indicators	Helmholtz coil calibration
Reversibility (< 10 kV/m indicators)	Full return to CC-0 after field removal	Field cycle test
UV stability (YInMn reference tile)	ΔE* < 1.0 after 1,000 kJ/m^2 UV	ISO 4892-2 accelerated

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11. References and Standards

- NOAA Space Weather Prediction Center, Schumann Resonance Background Data Revision 2025
- NOAA/SWPC Geomagnetic Storm Scale (G1–G5), Kp-index correlation tables
- ASTM D257: Standard Test Methods for DC Resistance or Conductance of Insulating Materials
- ISO 4892-2: Plastics — Methods of Exposure to Laboratory Light Sources
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- Pozar, D.M., *Microwave Engineering*, 4th ed. — ELF passive resonator design principles

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"Electronics cannot survive what they are designed to detect. The only instrument that will reliably tell you the severity of a Carrington event is one that was never electronic to begin with. Color is permanent. Chromism is honest. The storm cannot blind a detector that sees with chemistry rather than circuits."

— CSM Engineering Design Directive, Aegis Home Series

Document prepared by: Carrington Storm Motors / Safe Pod Engineering Company
CSMFAB Publication 021 | Version 2.0 | June 2026